

Searching Tables.

Creating Pre-filled Tables

A B C D E F G H I J K L

```
01 LetterTable.  
  02 TableValues.  
    03 FILLER PIC X(13)  
      VALUE "ABCDEFGHIJKLM".  
    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".
```

Creating Pre-filled Tables

A	B	C	D	E	F	G	H	I	J	K	L
---	---	---	---	---	---	---	---	---	---	---	---

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  02 TableValues.  
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    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".  
  
02 FILLER REDEFINES TableValues.  
  03 Letter PIC X OCCURS 26 TIMES.
```

Searching a Table

A	B	C	D	E	F	G	H	I	J	K	L
1	2	3	4	5	6	7	8	9	10	11	12

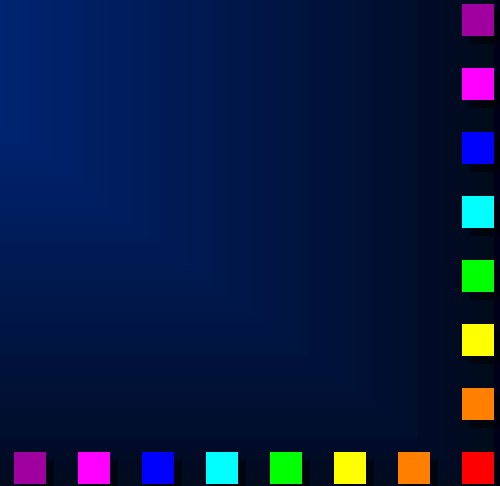
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      VALUE "NOPQRSTUVWXYZ".  
  02 FILLER REDEFINES TableValues.  
    03 Letter PIC X OCCURS 26 TIMES.
```

```
PERFORM VARYING Idx FROM 1 BY 1 UNTIL  
  LetterIn EQUAL TO Letter(Idc)  
END-PERFORM.  
DISPLAY LetterIn, "is in position ", Idc.
```

Search Syntax

SEARCH TableName $\left[\underline{\text{VARYING}} \begin{Bmatrix} \text{Identifier} \\ \text{IndexName} \end{Bmatrix} \right]$
[AT END StatementBlock]
 $\left\{ \underline{\text{WHEN}} \text{ Condition} \begin{Bmatrix} \text{Statementblock} \\ \underline{\text{NEXT SENTENCE}} \end{Bmatrix} \right\} \dots$
END – SEARCH

OCCURS TableSize TIMES
[INDEXED BY {IndexName}...]

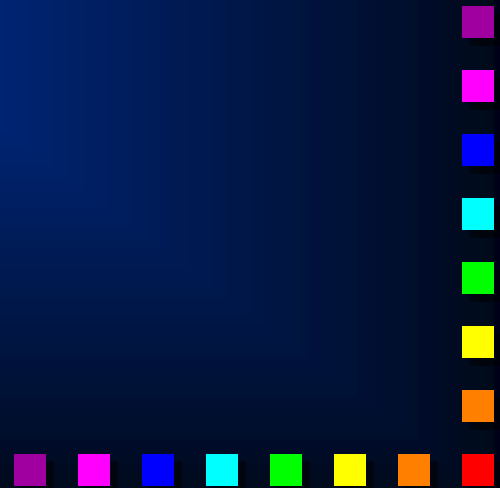


SET Syntax

SET $\left\{ \begin{array}{l} \text{IndexName} \\ \text{Identifier} \end{array} \right\} \dots \text{TO} \left\{ \begin{array}{l} \text{IndexName} \\ \text{Identifier} \\ \text{Integer} \end{array} \right\}$

SET $\{ \text{IndexName} \} \dots \left\{ \begin{array}{l} \text{UP} \\ \text{DOWN} \end{array} \right\} \text{BY} \left\{ \begin{array}{l} \text{Identifier} \\ \text{Integer} \end{array} \right\}$

SET $\{ \text{ConditionName} \} \dots \text{TO TRUE}$



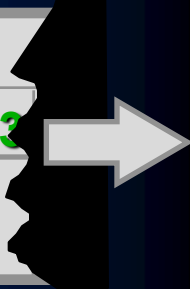
Searching a Table

A	B	C	D	E	F	G	H	I	J	K	L	
1	2	3	4	5	6	7	8	9	10	11	12	

```
01 LetterTable.  
  02 TableValues.  
    03 FILLER PIC X(13)  
      VALUE "ABCDEFGHIIJKLM".  
    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".  
  02 FILLER REDEFINES TableValues.  
    03 Letter PIC X OCCURS 26 TIMES  
      INDEXED BY LetterIdx.  
  
SET LetterIdx TO 1.  
SEARCH Letter  
  AT END DISPLAY "Letter not found!"  
  WHEN Letter(LetterIdx) = LetterIn  
    DISPLAY LetterIn, "is in position ", Idx  
END-SEARCH.
```

Searching a Two Dimension Table.

1								2								3										
1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3



```
01 TimeTable.
```

```
02 Day OCCURS 5 TIMES INDEXED BY DayIdx.
```

```
03 Hours OCCURS 8 TIMES INDEXED BY HourIdx.
```

```
04 Item          PIC X(10).
```

```
04 Location      PIC X(10).
```

```
SET DayIdx TO 0.
```

```
PERFORM UNTIL MeetingFound OR DayIdx > 5
```

```
SET DayIdx UP BY 1
```

```
SET HourIdx TO 1
```

```
SEARCH Hours WHEN MeetingType = Item(DayIdx, HourIdx)
```

```
SET MeetingFound TO TRUE
```

```
DISPLAY MeetingType " on " DayIdx " at " HourIdx
```

```
END-SEARCH
```

```
END-PERFORM.
```


Search All Syntax.

SEARCH ALL TableName [AT END StatementBlock]

WHEN $\left\{ \begin{array}{l} \text{ElementIdentifier} \left\{ \begin{array}{l} \text{IS } \underline{\text{EQUAL TO}} \\ \text{IS } = \end{array} \right\} \left\{ \begin{array}{l} \text{Identifier} \\ \text{Literal} \\ \text{ArithExpression} \end{array} \right\} \\ \text{ConditionName} \end{array} \right\}$

$\left[\begin{array}{l} \left\{ \begin{array}{l} \underline{\text{AND}} \left\{ \begin{array}{l} \text{ElementIdentifier} \left\{ \begin{array}{l} \text{IS } \underline{\text{EQUAL TO}} \\ \text{IS } = \end{array} \right\} \left\{ \begin{array}{l} \text{Identifier} \\ \text{Literal} \\ \text{ArithExpression} \end{array} \right\} \\ \text{ConditionName} \end{array} \right\} \end{array} \right]$

$\left\{ \begin{array}{l} \text{StatementBlock} \\ \text{NEXT SENTENCE} \end{array} \right\}$

END – SEARCH

OCCURS TableSize TIMES

$\left[\left\{ \begin{array}{l} \underline{\text{ASCENDING}} \\ \underline{\text{DESCENDING}} \end{array} \right\} \text{KEY IS } \{ \text{ElementIdentifier} \} \dots \right] \dots$

$\left[\underline{\text{INDEXED BY}} \{ \text{IndexName} \} \dots \right]$

Using the Search All

A	B	C	D	E	F	G	H	I	J	K	L	
1	2	3	4	5	6	7	8	9	10	11	12	

```
01 LetterTable.  
  02 TableValues.  
    03 FILLER PIC X(13)  
      VALUE "ABCDEFGHIJKLM".  
    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".  
  02 FILLER REDEFINES TableValues.  
    03 Letter PIC X OCCURS 26 TIMES  
      ASCENDING KEY IS Letter  
      INDEXED BY LetterIdx.  
  
SEARCH ALL Letter  
  WHEN Letter(LetterIdx) = LetterIn  
    DISPLAY LetterIn, "is in position ", Idx  
END-SEARCH.
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

1

Upper

26

Middle

13

=

Letter(Middle)

M

ALGORITHM.

$\text{Middle} = (\text{Lower} + \text{Upper}) / 2$

CASE TRUE

WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1

WHEN Letter(Middle) > "Q" THEN Upper = Middle - 1

WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE

WHEN Lower > Upper THEN ItemNotInTable TO TRUE

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

14

Upper

26

Middle

13

=

Letter(Middle)

M

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Middle = (Lower + Upper) / 2

CASE TRUE

WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

14

Upper

26

Middle

20

=

Letter(Middle)

T

ALGORITHM.

Middle = (Lower + Upper) / 2

CASE TRUE

WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1

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How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

14

Upper

19

Middle

20

=

Letter(Middle)

T

ALGORITHM.

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle		Letter(Middle)
14	19	16	=	P

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

17

Upper

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Middle

16

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Lower

17

Upper

19

Middle

18

=

Letter(Middle)

R

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A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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Lower

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Middle

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Letter(Middle)

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`Middle = (Lower + Upper) / 2`

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`WHEN Lower > Upper THEN ItemNotInTable TO TRUE`

Search All Example.

```
01 StateTable.  
  02 StateValues.  
    03 FILLER PIC X(20) VALUE ??????????????  
        Post Codes and Names  
    03 FILLER PIC X(20) VALUE ??????????????
```

```
02 FILLER REDEFINES StateValues.  
  03 States OCCURS 50 TIMES  
      ASCENDING KEY IS StateName  
      INDEXED BY StateIdx.  
    04 PostCode      PIC X(6) .  
    04 StateName     PIC X(14) .
```

```
SEARCH ALL States  
  AT END DISPLAY "State not found"  
  WHEN StateName(StateIdx) = InputName  
    MOVE PostCode(StateIdx) TO PrintPostCode  
END-SEARCH.
```